

Northwind Traders Sales & Operations Analysis – Power BI Dashboard

1. Project Overview

This project was completed as part of a Maven Analytics challenge, where I worked as a BI Developer for **Northwind Traders**, a global import and export company specializing in gourmet food products.

The objective was to design an executive-level KPI dashboard to provide quick insights into sales performance, product trends, customer behavior, and shipping efficiency.

2. Business Problem / Objectives

The executive team required a centralized dashboard to monitor key business areas and support strategic decisions.

The analysis aimed to answer:

- Are there any noticeable sales trends over time?
- Which products perform best and worst?
- Who are the key customers driving revenue?
- Are shipping costs consistent across providers?

3. Dataset Description

- **Source:** Maven Analytics (Northwind Dataset)
- **Data Structure:** 7 relational tables

Key Data Tables:

- Customers
- Orders
- Order Details
- Products
- Employees
- Shippers
- Categories

4. Data Cleaning & Preparation (Power Query)

- Data was **well-structured**, requiring minimal cleaning
- Performed transformations using **Power Query**:

- Data type corrections
- Minor data validation and formatting

5. Data Modeling & DAX

Data Modeling:

- Established relationships between all 7 tables
- Built a structured data model for efficient analysis

Key Measures Created:

- AOV (Average Order Value)
- OFT (Order Fulfillment Time)
- OTDR (On-Time Delivery Rate)
- SCPO (Shipment Cost per Order)

Additional Metrics:

- Total Sales & Sales Growth
- Order Count
- Shipping Cost Analysis
- Employee Performance Metrics

6. Dashboard Overview

An interactive Power BI dashboard was created to provide insights across multiple business areas.

Key Analysis Areas:

1. Sales Performance

- Sales trends over time
- Growth analysis

2. Product Performance

- Best-selling products
- Worst-performing products
- Orders vs revenue comparison

3. Customer Analysis

- Key customers by:
 - Sales contribution
 - Market share
 - Growth trends

4. Shipping & Logistics

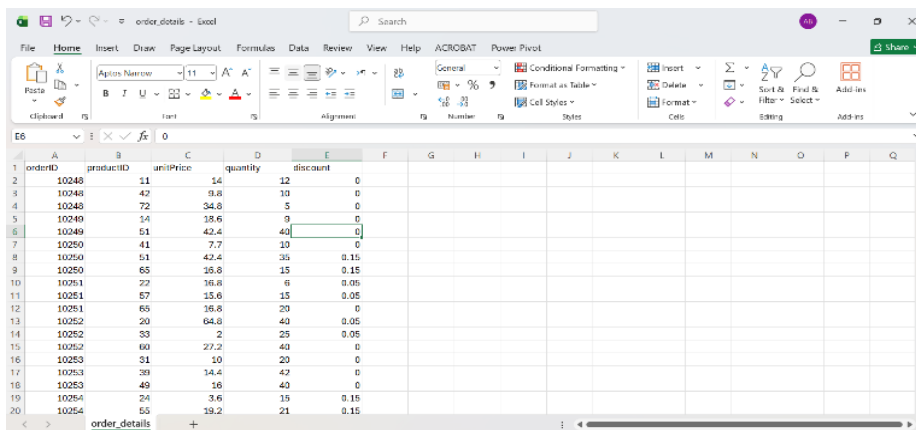
- Shipping cost comparison
- Delivery performance
- Consistency analysis across providers

5. Employee Performance

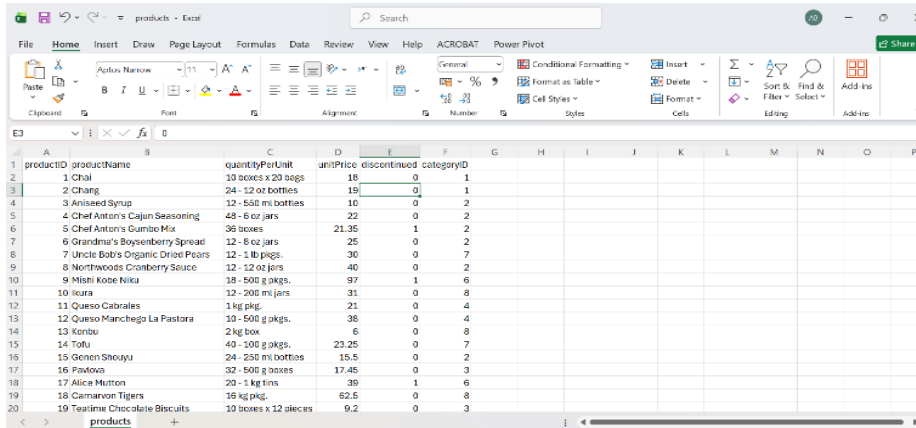
- Sales by employee
- On-time delivery rate
- Order fulfillment efficiency

7. Dashboard & Data Screenshots

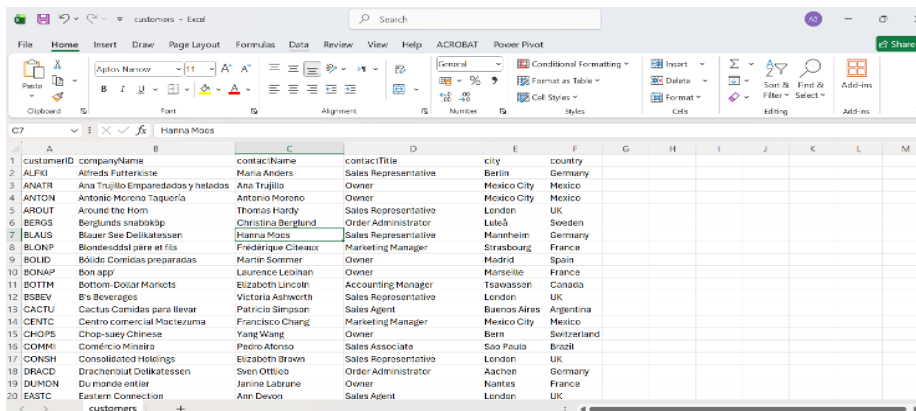
Figure 1: CSV files (Data sources)



orderID	productID	unitPrice	quantity	discount
10248	11	34	12	0
10248	42	9.8	10	0
10248	72	34.8	5	0
10249	54	18.6	9	0
10249	51	42.4	40	0
10250	41	7.7	10	0
10250	51	42.4	35	0.15
10250	65	16.8	15	0.15
10251	22	16.8	6	0.06
10251	67	15.6	15	0.05
10251	65	16.8	20	0
10252	20	64.8	40	0.05
10252	33	2	25	0.05
10252	60	27.2	40	0
10253	51	10	20	0
10253	39	14.4	42	0
10253	49	16	40	0
10254	24	3.6	15	0.15
10254	55	19.2	21	0.15



productID	productName	quantityPerUnit	unitPrice	discontinued	categoryID
1	Chai	10 boxes x 20 bags	18	0	1
2	Chang	24 - 12 oz bottles	19	0	1
3	Aniseed Syrup	12 - 550 ml bottles	10	0	2
4	Chef Anton's Cajun Seasoning	48 - 6 oz jars	22	0	2
5	Chef Anton's Gumbo Mix	36 boxes	21.35	1	2
6	Grandma's Boysenberry Spread	12 - 8 oz jars	25	0	2
7	Uncle Bob's Organic Dried Pears	12 - 1 lb pgs.	30	0	7
8	Northwoods Cranberry Sauce	12 - 12 oz jars	40	0	2
9	Mishi Kobe Niku	18 - 500 g pkgs.	97	1	6
10	Ikura	12 - 200 ml jars	31	0	8
11	Queso Cabrales	1 kg pkg.	21	0	4
12	Queso Manchego La Pastora	10 - 500 g pkgs.	38	0	4
13	Konbu	2 kg box	6	0	8
14	Tofu	40 - 100 g pkgs.	23.25	0	7
15	Genen Shouyu	24 - 250 ml bottles	15.5	0	2
16	Pastusa	32 - 500 g boxes	17.45	0	3
17	Alice Mutton	20 - 1 kg tins	39	1	6
18	Camaron Tigers	16 kg pkg.	62.5	0	8
19	Teatime Chocolate Biscuits	10 boxes x 12 pieces	9.2	0	3



customerID	companyName	contactName	contactTitle	city	country
ALFKI	Alfreds Futterkiste	Maria Anders	Sales Representative	Berlin	Germany
ANATR	Ana Trujillo Emparedados y helados	Ana Trujillo	Owner	Mexico City	Mexico
ANTON	Antonio Moreno Taquería	Antonio Moreno	Owner	Mexico City	Mexico
AROUT	Around the Horn	Thomas Hardy	Sales Representative	London	UK
BERGS	Berglunds snabbköp	Christina Berglund	Order Administrator	Luleå	Sweden
BLAUS	Bauer Stehls Delikatessen	Hanna Moos	Sales Representative	Mannheim	Germany
BONAP	Bonaparte	Frédérique Citeaux	Marketing Manager	Strasbourg	France
BOLID	Bólido Comidas preparadas	Martin Sommer	Owner	Madrid	Spain
BONAP	Bon app	Laurence Leblanc	Owner	Marseille	France
BOTM	Bottom-Dollar Markets	Elizabeth Lincoln	Accounting Manager	Tampa	USA
BSBEV	B's Beverages	Victoria Ashworth	Sales Representative	London	UK
CACTU	Cactus Comidas para llevar	Patricio Simpson	Sales Agent	Buenos Aires	Argentina
CENTC	Centro comercial Maciezuma	Francisco Chang	Marketing Manager	Mexico City	Mexico
CHOPS	Chop-suey Chinese	Yang Wang	Owner	Bern	Switzerland
COMM	Comércio Mineiro	Pedro Afonso	Sales Associate	Sao Paulo	Brazil
CONSH	Consolidated Holdings	Elizabeth Brwon	Sales Representative	London	UK
DRACD	Drachenstall Delikatessen	Sven Ottlieb	Order Administrator	Aachen	Germany
DUMON	Du monde entier	Janina Labrune	Owner	Nantes	France
EASTC	Eastern Connection	Ann Devon	Sales Agent	London	UK

orderID	customerID	employeeID	orderDate	requiredDate	shippedDate	shipperID	freight
10248	VINET	5	04/07/2013	01/08/2013	16/07/2013	3	52.38
10249	TOMSP	6	05/07/2013	16/08/2013	10/07/2013	1	11.61
10250	HANAR	4	08/07/2013	09/08/2013	12/07/2013	2	65.83
10251	VICTE	3	08/07/2013	05/08/2013	15/07/2013	1	41.34
10252	SUPMD	4	08/07/2013	09/08/2013	11/07/2013	2	51.3
10253	HANAR	3	10/07/2013	24/07/2013	16/07/2013	2	58.17
10254	CHOPS	5	11/07/2013	08/08/2013	23/07/2013	2	22.86
10255	RICSU	9	12/07/2013	09/08/2013	15/07/2013	3	148.33
10256	WELLI	3	15/07/2013	12/08/2013	17/07/2013	2	13.97
10257	HILAA	4	16/07/2013	13/08/2013	22/07/2013	3	81.81
10258	ERNHS	1	17/07/2013	14/08/2013	23/07/2013	1	140.51
10259	CENTC	4	18/07/2013	15/08/2013	29/07/2013	3	3.25
10260	OTTIK	4	18/07/2013	16/08/2013	29/07/2013	1	55.09
10261	QUEDE	4	18/07/2013	16/08/2013	30/07/2013	2	3.05
10262	RATTC	8	22/07/2013	19/08/2013	25/07/2013	3	48.29
10263	HRNHS	9	23/07/2013	20/08/2013	31/07/2013	3	146.06
10264	FOLKO	6	24/07/2013	21/08/2013	23/08/2013	3	3.67
10265	BLOMP	2	25/07/2013	22/08/2013	12/08/2013	1	55.28
10266	WARTH	3	26/07/2013	09/08/2013	31/07/2013	3	25.73

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Figure 2: Data Transformation (Power Query)

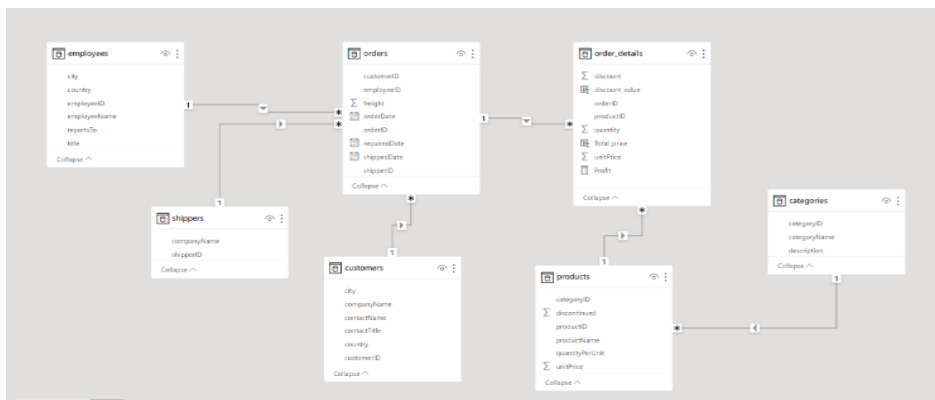


Figure 3: Data Modeling

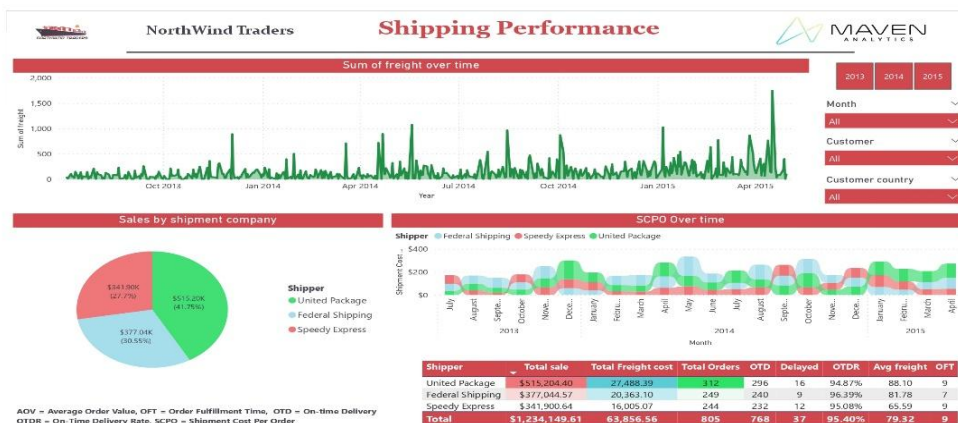


Figure 4: Dashboard

8. Key Insights & Recommendations

Sales Trends

- Sales showed a **significant upward trend**, with **~296% growth** between July 2013 and April 2015

Recommendation:

Capitalize on growth momentum by identifying high-performing periods and aligning marketing and inventory strategies accordingly.

Product Performance

- Clear distinction between **top-performing and underperforming products** based on sales and order volume

Recommendation:

Focus on promoting high-performing products while reviewing pricing, demand, or discontinuation strategies for low-performing items.

Customer Analysis

- Identified key customers contributing the highest revenue and market share
- Some customers showed **low current sales but high growth potential**

Recommendation:

Strengthen relationships with top customers while targeting high-growth customers for future expansion.

Shipping Performance

- Among three shipping providers, **Speedy Express demonstrated the most consistent performance** based on lower cost variability (standard deviation)

Recommendation:

Prioritize reliable shipping partners to ensure cost stability and improve delivery performance.

Operational Metrics

- Metrics such as **On-Time Delivery Rate, Order Fulfillment Time, and Shipment Cost per Order** provided visibility into operational efficiency

Recommendation:

Continuously monitor operational KPIs to improve delivery timelines and reduce logistics costs.

Employee Performance

- Employee-level analysis highlighted differences in:
 - Sales performance

- Delivery efficiency
- Fulfillment time

Recommendation:

Use performance insights for training, workload balancing, and recognizing top-performing employees.

9. Tools & Technologies Used

- Power BI – Dashboard development & DAX
- Power Query – Data transformation
- Excel / CSV – Data source

10. Key Skills Demonstrated

- Data modeling & relational analysis
- KPI development using DAX
- Dashboard design & storytelling
- Business performance analysis
- Operational and logistics insights